

Ex. 61. A train starts full of passengers. At the first station, it drops one-third of the passengers and takes 280 more. At the second station, it drops one-half of the new total and takes 12 more. On arriving at the third station, it is found to have 248 passengers. Find the number of passengers in the beginning.

Sol. Let the number of passengers in the beginning be x .

$$\text{After 1st station, number of passengers} = \left(x - \frac{x}{3}\right) + 280 = \left(\frac{2x}{3} + 280\right).$$

$$\text{After 2nd station, number of passengers} = \frac{1}{2} \left(\frac{2x}{3} + 280\right) + 12.$$

$$\therefore \frac{1}{2} \left(\frac{2x}{3} + 280\right) + 12 = 248 \Leftrightarrow \frac{2x}{3} + 280 = 2 \times 236 \Leftrightarrow \frac{2x}{3} = 192$$

$$\Leftrightarrow x = \left(192 \times \frac{3}{2}\right) = 288.$$

$\therefore$ Required number of passengers = 288.

Ex. 62. Arun and Sajal are friends. Each has some money. If Arun gives ₹ 30 to Sajal, then Sajal will have twice the money left with Arun. But, if Sajal gives ₹ 10 to Arun, then Arun will have thrice as much as is left with Sajal. How much money does each have?

Sol. Suppose Arun has ₹ x and Sajal has ₹ y . Then,

$$2(x - 30) = y + 30 \Rightarrow 2x - y = 90 \quad \dots(i)$$

$$\text{and } x + 10 = 3(y - 10) \Rightarrow x - 3y = -40 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 62$ and $y = 34$.

$\therefore$ Arun has ₹ 62 and Sajal has ₹ 34.

Ex. 63. An executive goes on a business trip. His daily schedule has a definite pattern. If he is busy with a meeting in the morning, he is free in the afternoon. When he returns, he realises that he attended 15 meetings altogether. There were 12 free mornings and 13 free afternoons. What was the duration of his trip?

Sol. Let M and A represent the number of busy mornings and busy afternoons respectively.

Then, total number of mornings in the trip = $12 + M$.

And, total number of afternoons in the trip = $13 + A$.

$$\text{Clearly, } 12 + M = 13 + A \text{ or } M - A = 1 \quad \dots(i)$$

Total number of meetings = $M + A$

$$\text{So, } M + A = 15 \quad \dots(ii)$$

Adding (i) and (ii), we get: $2M = 16$ or $M = 8$.

Hence, duration of the trip = $(12 + 8) = 20$ days.

Ex. 64. A man received a cheque. The rupees had been transposed for paise and vice-versa. After spending 5 rupees 42 paise, he discovered that he now had exactly six times the value of the correct cheque amount. What amount should he have received? (M.A.T., 2005)

Sol. Let the amount received be x rupees and y paise.

Then, amount written on cheque = $(100x + y)$ paise.

Actual amount to be received = $(100y + x)$ paise.

$$\therefore (100x + y) - 542 = 6(100y + x)$$

$$\Leftrightarrow 100x + y - 542 = 600y + 6x \Leftrightarrow 94x - 599y = 542$$

$$\Leftrightarrow 94x = 542 + 599y$$

$$\Leftrightarrow x = \frac{542 + 599y}{94} = \frac{470 + 470y + 72 + 129y}{94} = 5(1 + y) + \frac{72 + 129y}{94}.$$

For an integral value of x , $(72 + 129y)$ must be divisible by 94. This happens when $y = 6$.

$$\therefore x = 5(1 + 6) + \frac{72 + 129 \times 6}{94} = 35 + 9 = 44.$$

Hence, the man should have received ₹ 6.44.